

Week 4 Report: Data Preprocessing and Exploratory Data Analysis

Project: Determining the Role of Big Data Applications in Fraud Detection and Financial Security in Financial Institutions of Ireland

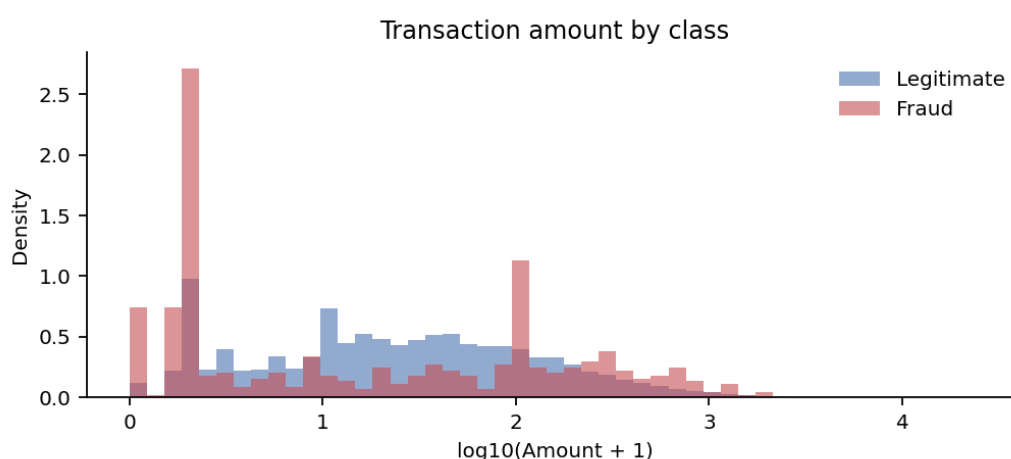
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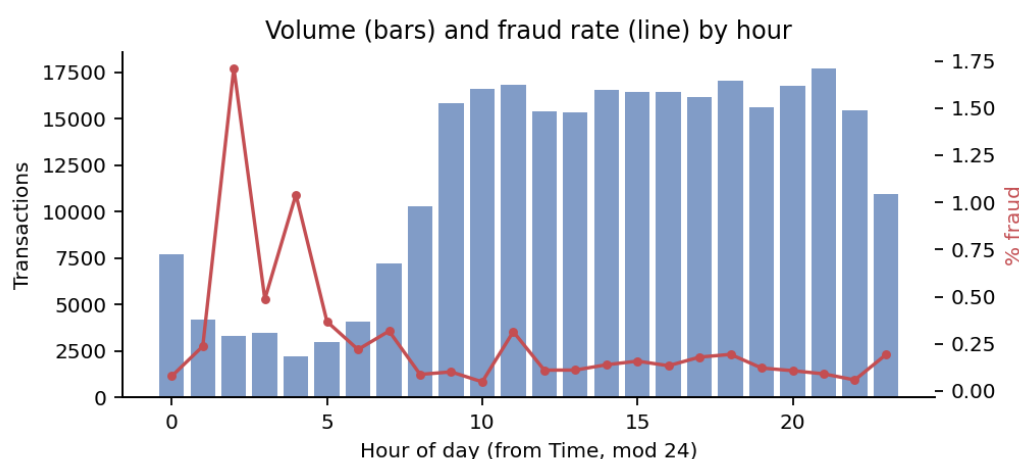
What I did this week

This week I finished preprocessing and started the exploratory analysis. Time, Amount and the new Hour feature are standardised, and the scaler is fitted on training data only so no test information is used. V1 to V28 are already centred PCA components, so they are left as they are.

The exploration confirmed three patterns from Week 2. Fraud amounts have a lower median but a longer tail than legitimate ones, fraud is about three to four times more common between midnight and 6am, and V17, V14, V12 and V10 are the components most correlated with fraud.



Transaction amounts by class (log scale).



Transactions and fraud rate by hour of day.

Next week

Build the model pipeline and run a first comparison of the three models.